Measurements on a test machine

Measurement chapter, draft

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Chapter 3

Measurements on a test machine

The theory of Chapter 4 takes a few numbers per function as given: how long a snapshot restore takes, how much warm-up is left after it, how much parallel restores slow each other down, and how a start-up responds to CPU. The simulator used estimates for all of them. This chapter measures them on real hardware, and answers two questions raised in review: whether a snapshot mechanism is worth its cost against what the JDK itself now offers, and whether the premise of the CPU plan holds.

3.1 Method

Machine. A virtual machine with 8 Intel Xeon Gold 6248R cores (2 sockets, one thread per core), 31 GB of memory, Ubuntu 24.04 with Linux 6.8, cgroup v2, no CPU steal time (checked with `vmstat`), on VMware. Software: Docker 29.8, CRIU 4.2.1, the Zulu CRaC build of OpenJDK 21.0.12, and OpenJDK 25.0.4. CPU quotas are set with the cgroup `cpu.max` file; CPU-seconds and throttled periods are read from `cpu.stat`.

Workload. A Spring Boot function (web starter, Jackson, Hibernate Validator) whose handler deserialises, validates, aggregates and serialises a list of 400 orders; the service is called directly, so the measurement is of framework start-up and JIT warm-up, not of socket handling. Every run uses the same serving shape: `-XX:ActiveProcessorCount=1`, the serial collector, a 256–512 MB heap.

Snapshots. Snapshots are taken with CRaC (Coordinated Restore at Checkpoint): the function calls the checkpoint itself after K warm-up requests (depth K), and a restore resumes it at that point. CRaC has two engines. The Zulu build checkpoints with Azul's *Warp* engine unless told otherwise; CRIU is selected with `-XX:CRaCEngine=criu`. Most measurements below used Warp; Section 3.3 compares the two engines directly and shows that the conclusions hold for CRIU.

Metrics. *Time to first response* (TTFR): from the launch of the process, or of the restore, to the moment the first request's response is complete, measured outside the process (the function prints the wall-clock time of its first response). After the first response every run serves the same 200 requests. C is the steady latency (median of requests 150–200 of the most warmed process); the *residual warm-up* is $R = \sum_i \max(0, \ell_i - C)$ over the first request and the 200 after it, where ℓ_i is the latency of request i; for a cold start this is the warm-up B.

Rules. Every test followed the same rules. Its prediction was written and committed before the run. It has a control whose answer is known in advance, run first. Each cell has $n \geq 20$ repetitions in random order. Raw data has one row per run and is never edited. Medians are

reported with 95% bootstrap confidence intervals (10,000 resamples). A falsified prediction is reported as a result. Table 3.5 lists every prediction and its outcome.

3.2 Is a snapshot worth it?

JDK 24 and 25 ship a competitor that needs no snapshot: the ahead-of-time cache of Project Leyden [4, 3, 5]. A training run records the classes it loaded and linked, and the profiles of the methods it ran; later starts reuse them. It needs no special privileges and cannot clone secrets. If it came close to a snapshot restore, a snapshot mechanism would not be worth its operational cost. The registered decision rule: a snapshot is justified if it reaches the first response at least 3× sooner than the AOT cache trained with the same requests, at both CPU levels.

Table 3.1: Start modes of the same function: time to first response (median, $n = 20$) and CPU per start. The AOT cache and the restore at depth 50 saw the same 50 training requests.

start mode	0.25 vCPU	1 vCPU	CPU-s
cold start, JDK 21 / JDK 25	16.9 / 17.6 s	3.97 / 4.08 s	5.4–5.7
JIT capped at C1 (`-XX:TieredStopAtLevel=1`)	9.8 s	2.32 s	3.0
JDK 25 AOT cache, trained with start-up only	9.5 s	2.26 s	3.6–3.7
JDK 25 AOT cache, trained with 50 requests	8.7 s	2.05 s	3.0–3.3
snapshot restore after start-up ($K = 0$, Warp)	2.09 s	0.50 s	2.0
snapshot restore at $K = 50$, Warp	**0.29 s**	**0.08 s**	1.2
snapshot restore at $K = 50$, CRIU	**0.38 s**	**0.08 s**	1.0

Table 3.1 shows the result. The AOT cache roughly halves a cold start (0.54–0.55×), as published for Spring applications. A snapshot at depth 50 answers its first request **25–30× sooner** than the AOT cache with the same training (95% confidence intervals of the ratio: $[29.2, 33.1]$ at 0.25 vCPU, $[23.3, 31.7]$ at 1 vCPU; with the CRIU engine ≈ 23–24×), and uses about 2.5× less CPU per start. Even the snapshot point of current restore-on-demand systems ($K = 0$) is 4.5× ahead. The control passed: cold starts on JDK 21 and 25 differ by 3–4%, so the comparison is not an artefact of the JDK version.

One prediction failed. After a depth-50 restore, the residual warm-up over the next 200 requests is *not* smaller than after the AOT cache (720 vs 652 ms at 1 vCPU, overlapping intervals). A snapshot's advantage is the first response; the JIT still has work to do afterwards (Section 3.5).

3.3 The two engines

Table 3.2: Warp vs CRIU on the same function (time to first response, median [95% CI], $n = 20$).

vCPU	K	Warp	CRIU	difference
1	0	474 [463, 483] ms	492 [482, 502] ms	+18 ms
1	50	67 [61, 72] ms	84 [80, 91] ms	+17 ms
0.25	0	1986 [1967, 2035] ms	2076 [2066, 2131] ms	+90 ms
0.25	50	284 [280, 291] ms	375 [301, 385] ms	+91 ms

All 160 restores and 16 checkpoints succeeded with both engines. CRIU is 17–91 ms slower per restore (Table 3.2), with the same CPU-seconds and similar image sizes (78–98 vs 79–93 MB). The prediction, "a few hundred milliseconds slower", was falsified in CRIU's favour. The first request after the restore takes the same time with either engine, so the engines differ only in the restore itself. The design does not depend on the engine.

3.4 Parallel restores

Look-ahead restores several stages at once. Theorem 4.7 of Chapter 4 bounds the gain by $(1 - \beta)$, where β measures how much N simultaneous restores slow the slowest one: $\beta(N) = (r_{\max}(N)/r(1) - 1)/(N - 1)$; $\beta = 1$ means the restores run one after another.

The snapshot (Warp, $K = 50$) was restored $N \in \{1, 2, 4, 8\}$ times at once, each copy in its own Docker container with its own CPU quota, 20 repetitions, about 2,400 restores in total, none failed. Through `docker run`, $\beta(4) = 0.21$ $[0.17, 0.27]$ at 0.25 vCPU and 0.53 $[0.35, 0.67]$ at 1 vCPU. A control that starts the same number of containers running an empty command slows down exactly as much: Docker creates containers one after another, about 250 ms each [2]. Restoring into containers created beforehand (as OpenWhisk's prewarmed "stem cell" containers are) removes it:

restore into an existing container	$N = 1$	2	4	8	$\beta(4)$ [95% CI]
0.25 vCPU	380 ms	418	429	456	0.04 [0.01, 0.08]
1 vCPU	101 ms	106	122	201	0.07 [0.03, 0.09]

The restores themselves barely compete: eight at once cost 76–100 ms more than one. The serial control (each copy started when the previous one answered) gave $\beta = 1.04$–1.14 at 0.25 vCPU, as it must, but 1.23–1.63 at 1 vCPU, because each copy's remaining work overlapped the next copy's start; the result for pre-created containers does not depend on this control. **Design consequence:** the platform should restore into existing containers, not create a container per restore.

3.5 The function's profile against snapshot depth

Snapshots were taken after $K \in \{0, 1, 5, 20, 50, 100, 200\}$ requests (4 images each, built at 1 vCPU) and restored at 0.25, 0.5, 1, 2 and 4 vCPU, 20 times each, interleaved with cold starts and with processes that were never checkpointed: 1,280 runs, none failed.

Table 3.3: The profile (Warp; ms, median of 20). TTFR / R for restores; B for cold starts.

vCPU	cold TTFR / B	$K = 0$	$K = 50$	$K = 200$	never checkpointed, $K = 200$: R
0.25	16 551 / 5 759	2 675 / 7 310	380 / 3 570	875 / 2 313	1 563
0.5	7 981 / 3 084	1 025 / 3 590	163 / 1 656	161 / 1 105	811
1	3 968 / 1 271	758 / 1 606	181 / 720	297 / 453	342
2	1 988 / 628	267 / 753	61 / 280	60 / 165	93
4	1 798 / 561	248 / 577	59 / 229	58 / 112	69

Image size is flat in K: 82–88 MB at every depth (the earlier cost analysis assumed 90 MB).

Restore time is flat in K for $K \geq 5$: about 60 ms at 2 and 4 vCPU and 67–84 ms at 1 vCPU (Table 3.2). The theory's examples used 650 ms; a deep restore is an order of magnitude cheaper.

Residual warm-up falls with depth and does not level off by $K = 200$ (at 1 vCPU: 1 606, 720, 453 ms at $K = 0$, 50, 200); the predicted 63 ms at $K = 50$ (from a lighter function) was off by 11×. The JIT is still compiling at $K = 200$: compile time at the checkpoint grows from 4.5 to 6.7 s and the code cache from 7.6 to 9.4 MB between $K = 0$ and 200. Theorem 4.23's stopping rule (stop when compilation stops) therefore did not trigger within 200 requests on this function.

An anomaly. At 0.25 and 1 vCPU, deep restores in this experiment were bimodal: at 1 vCPU and $K = 50$, ten runs took 61–88 ms and ten 274–386 ms. The same images restored in 59 ms when the runs were not interleaved with cold starts, and the dedicated engine test measured 67–84 ms unimodally. It is not eviction from the page cache (the cgroup recorded no refaults), not disk (a restore from disk takes about 505 ms), and not one image (all four show both modes). Its cause is open; the restore times quoted above are taken from the experiments without it.

Disk. With the page cache dropped first, a depth-50 restore takes 505 ms at 1 vCPU and 1 094 ms at 0.25 vCPU, against 67 and 284 ms from memory: reading an 88 MB image costs more than the restore. An invoker should keep its images in memory.

3.6 Warm-up lost to the checkpoint, and re-warming

A restored process has more warm-up left than the same process never checkpointed. The loss $L(K) = R_{\text{restored}} - R_{\text{never checkpointed}}$ at $K = 50$ is 955 ms at 0.25 vCPU, 163 ms at 1 vCPU and indistinguishable from zero at 2–4 vCPU (several intervals include zero; $K = 200$ is clearly positive up to 2 vCPU). The loss is larger where CPU is scarce, as predicted. A restored sandbox that serves 50 more test requests while it waits for its input (*re-warming*) ends with *less* residual than the never-checkpointed process (1 762 vs 2 615 ms at 0.25 vCPU; 337 vs 556 ms at 1 vCPU): re-warming does not just recover the loss, it deepens the snapshot. It costs 1.3 s of waiting time at 0.25 vCPU and 0.3 s at 1 vCPU.

3.7 CPU during start-up

The CPU plan (Section 4.9) rests on assumption B1 (Table 4.4): a start-up finishes in proportion to the CPU it gets, at the same CPU-seconds.

Table 3.4: Speed-up of the time to first response relative to 0.25 vCPU, and CPU-seconds per start (median of 20).

mode	0.5	1	2	4 vCPU	CPU-s, 0.25 → 4 vCPU
cold start	2.07×	4.17×	8.32×	9.2×	5.40 → 5.16
restore, $K = 0$	2.61×	3.53×	10.0×	10.8×	2.11 → 1.85
restore, $K = 50$ (engine test)		4.2–4.5×			1.1 → 1.0

B1 holds, beyond one core (Table 3.4). A cold JVM start is 4.2× faster with four times the CPU and 8.3× faster with eight times, using the same CPU-seconds (within 6%) at every quota. It levels off only past 2 vCPU. A JVM told it has one processor still uses about two cores while it starts (collector, compiler and class-loading threads). The model's cap of one core per start-up is therefore too low; about two cores is right for this runtime.

Boosting only while starting. A start-up was given 1 or 2 vCPU and lowered to 0.25 vCPU the moment its first response was out (the change took effect within 1.4–4.8 ms). Boosting costs no extra CPU. For a deep restore the following requests were as fast as without the boost (1.71 vs 1.68 ms). For a cold start or a restore at $K = 0$ they were 17–52% slower: the boost brought the first response forward, so less compilation had happened by then, and the rest of the warm-up moved into the following requests. **The boost buys the first response, not the warm-up**; it should last until the warm-up is done.

One core, several start-ups. Two or four depth-50 restores shared one core, either split equally or one at a time at the full core (Theorem 4.29). One at a time made them ready 19–27% sooner on average (4 restores: 203 vs 279 ms) and the first one 1.9–3.6× sooner, at an 8–12% later last one, as the proposition predicts.

The cliff. Figure 3.1 shows the time to first response plus the residual warm-up against the quota. A cold start's warm-up grows 10.3× from 4 to 0.25 vCPU (an earlier measurement on a heavier warm-up gave 13–35×). The prediction that deep snapshots flatten this cliff was falsified as a ratio: the deeper the snapshot, the steeper the relative cliff (12.7× at $K = 0$, 20.6× at $K = 200$), because at 4 vCPU a deep restore leaves almost nothing. In absolute time it holds: a deep snapshot saves 19.1 s per start at 0.25 vCPU and 2.2 s at 4 vCPU. Snapshots save the most exactly where serverless functions run.

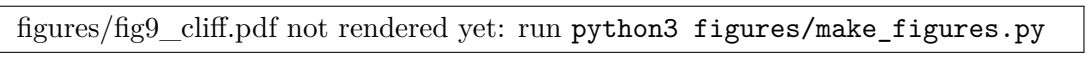


Figure 3.1: Time to first response plus residual warm-up against the CPU quota, for a cold start, a snapshot after start-up ($K = 0$) and a deep snapshot ($K = 200$); median of 20 runs per point, logarithmic axes.

3.8 Safety: what restored copies share

Copies restored from one image start with the same memory [1]. Ten copies were restored from one image and each reported hashes of values it had created at start-up. Without reset hooks, a `java.util.Random`, a secret token generated at start-up (even from a `SecureRandom`) and `Math.random()` returned the same value in all ten copies; the JDK's own `SecureRandom` and `UUID` sources were re-seeded on restore and differed. A CRaC resource hook that re-seeds the `Random` and regenerates the token fixed those two (ten distinct values), but **not `Math.random()`**, whose generator is private to the JDK and has no re-seed interface. A snapshot builder must therefore either refuse functions that call `Math.random()` before the checkpoint, or reset its generator by reflection; this is one concrete reason why not every function is safe to snapshot.

3.9 What the measurements change

- **The model's inputs.** Restore time r: 74–84 ms at 1 vCPU and 282–375 ms at 0.25 vCPU for a deep snapshot (was 650 ms). Contention $\beta \approx 0.05$ into existing containers (was an assumption). Residual warm-up after a restore: 720 ms over 200 requests at $K = 50$, 1 vCPU (was 63 ms). Start-up cap: about two cores (was one).
- **The weight shifts from timing to depth and CPU.** With $r \approx 80$ ms, restoring ahead hides about 80 ms per stage at 1 vCPU and about 300 ms at 0.25 vCPU. The residual warm-up and the CPU a start-up gets now matter more than when it starts.
- **The design.** Restore into existing containers; keep images in memory; boost a start-up until its warm-up is done; keep re-warming; choose the depth from the residual curve rather than from a compile counter that has not levelled off; exclude or patch functions that use `Math.random()` before the checkpoint.

3.10 Threats to validity

- **One function.** All numbers are for one Spring Boot function. The residual warm-up and the depth at which it levels off depend on the function; the qualitative results (snapshot vs

Table 3.5: Registered predictions and outcomes.

prediction	outcome
a deep snapshot reaches the first response $\geq 5\times$ sooner than the AOT cache	confirmed, 25–30×
residual warm-up after a deep restore $\ll$ after the AOT cache	falsified: similar
CRIU a few hundred ms slower than Warp	falsified: 17–91 ms
$\beta < 0.3$ for $N \leq 4$	confirmed into existing containers; through `docker run` only at 0.25 vCPU
Docker's own part of β: 0.1–0.3	falsified: 0.6–0.8
image size flat in K	confirmed, 82–88 MB
residual warm-up ≈ 63 ms at $K = 50$, levelling off	falsified: 720 ms, still falling at $K = 200$
restore from disk < 50 ms slower than from memory	falsified: 440–810 ms
$L(K) > 0$, larger at low CPU; re-warming recovers $\geq 30\%$	confirmed; recovers more than all of it
reset hooks make every value distinct	falsified for `Math.random()`
cold warm-up cliff 13–35×	falsified in size: 10.3×
deep snapshots flatten the cliff (ratio $< 5\times$)	falsified as a ratio (20.6×); holds in absolute time
start-up $\approx 4\times$ faster from 0.25 to 1 vCPU at the same CPU-seconds	confirmed (3.5–4.5×); continues to 2 cores
boosted start-ups serve later requests as fast	confirmed for deep restores; falsified for cold starts and $K = 0$

AOT cache, contention into existing containers, the shape of the speed-up) are the ones to generalise, and the next experiments add Python and ML stages.

- **A virtual machine.** The host is VMware with no measured steal time; the disk is virtual, so the disk tier's cost is specific to it.
- **Memory was not measured.** The cgroup's memory counter included the page cache it had accumulated, so the processes' own memory is unknown; image sizes are reported instead.
- **The bimodal restores** (Section 3.5) are unexplained.
- **Privileges.** Containers ran with `-privileged`; the minimal capability set, and CRIU inside pre-created containers, are left to the OpenWhisk integration.

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